

Harsh Vardhan Bhanot

(361) 742-6865 hvbhanot1@gmail.com linkedin.com/in/hvbhanot github.com/hvbhanot

Education

Bachelor of Science in Computer Science, Minor in Applied Mathematics

Texas A&M University–Corpus Christi

Expected 2026

Technical Skills

- **Programming Languages:** Python, Go, C++, Bash, SQL
- **ML & AI:** PyTorch, Hugging Face, scikit-learn, parameter-efficient fine-tuning
- **Scientific Computing:** Simulation analysis, experiment automation, reproducibility tooling
- **Systems:** Linux, macOS, Windows
- **Tools:** Git, REST APIs, JWT, JSON

Research Experience

Undergraduate Research Assistant — Computational Genetics & AI Systems 2025 – Present

Texas A&M University–Corpus Christi

- Conduct research using SLiM evolutionary simulation frameworks to model genetic evolution, mutation dynamics, and population-level fitness behavior.
- Develop Python-based analysis pipelines to parse SLiM outputs, track mutations across generations, and ensure reproducible experiment execution.
- Design lightweight agent-based workflows to automate simulation parameter sweeps and comparative outcome analysis.
- Fine-tune transformer-based models using Hugging Face for code understanding, experiment summarization, and research workflow support.
- Apply parameter-efficient fine-tuning techniques (e.g., adapters, LoRA-style methods) and evaluate model behavior on domain-specific code corpora.
- Maintain structured experiment logs, version-controlled codebases, and reproducible research artifacts.

Professional Experience

AI & ML Intern — DataEssenceAI

May 2024 – Aug 2024

Houston, TX

- Developed and evaluated machine learning models using PyTorch and OpenCV for market trend analysis.
- Performed hyperparameter tuning, reducing training time by 20% and improving predictive accuracy by 10%.
- Collaborated with data science and DevOps teams to support testing and deployment pipelines.

Leadership & Academic Service

Founder & Vice President — Islanders Research in AI

Jan 2023 – Present

- Founded and led a student research organization focused on artificial intelligence and machine learning.
- Organized workshops on ML fundamentals, research tooling, and reproducible experimentation.

Technical Projects

Clutch-Call — Sports Analytics & Prediction System

github.com/JustinSu11/clutch-call

- Contributed to a collaborative machine learning project analyzing high-pressure decision-making scenarios in sports.
- Implemented data preprocessing, feature engineering, and model evaluation pipelines.

Neural Network from Scratch — Iris Classification

- Implemented a multi-layer neural network using NumPy without deep learning frameworks.
- Achieved over 90% test accuracy using ReLU activations, softmax outputs, and gradient descent.